PROJECT SYNOPSIS & PROPOSAL

CAMPUSAI: AN INTELLIGENT AI-POWERED COLLEGE ASSISTANT CHATBOT

Bachelor of Business Administration (Computer Applications) — BBA (CA) / BCA

Project Title:	CampusAI: AI College Assistant Chatbot
Domain:	Artificial Intelligence & Web Application Development
Students:	Akash rana (ISTU25010120004) & Shreyash katke (ISTU25010120003)
Project Guide:	Prof. [Guide Name], Dept. of Computer Applications
Technologies:	Python 3, Flask, SQLite 3, SQLAlchemy, HTML5, CSS3, JavaScript, Gemini API

1. INTRODUCTION & PROBLEM CONTEXT

Collegiate administrative offices encounter hundreds of repetitive student inquiries every day concerning admissions, fees, examinations, course curriculums, and campus facilities. Manual resolution through office counters creates long queues, restricted working hours, and human fatigue. **CampusAI** solves this by delivering an automated, 24/7 web assistant that answers student questions instantaneously while enforcing a strict *Zero-Hallucination Policy* regarding institutional dates and fees.

2. PROJECT OBJECTIVES

- Provide 24/7 sub-second inquiry response via natural language processing.
- Implement a Dual-Tier response pipeline: deterministic FAQ token matching + context-bounded LLM fallback.
- Build an interactive FAQ directory with category filtering and real-time search.
- Engineer an authenticated Admin Dashboard for real-time CRUD management of FAQs.
- Maintain 100% factual accuracy by routing unknown questions to college office contacts.

3. SYSTEM ARCHITECTURE & KEY MODULES

- **Student Chat Module:** Conversational UI with question suggestion chips, typing indicator, and clear chat option.
- **Smart FAQ Engine:** Tokenizer with punctuation stripping, stop-word filtering, domain synonym expansion, and Jaccard scoring.
- **AI Service Layer:** Bounded Gemini LLM fallback with low temperature (0.2) and prompt engineering against hallucination.
- **Admin Governance Portal:** Protected by PBKDF2:SHA256 password hashing and session tokens for real-time FAQ management.
- **Database (3NF):** Normalized SQLite schema (`admins`, `faqs`, `chat_messages`, `query_logs`).

4. SYSTEM SPECIFICATIONS

Component	Minimum Requirement	Implemented Technology
Operating System	Windows 10 / macOS / Linux	Cross-platform (macOS / Linux)
Backend Framework	Python 3.8+	Python 3.10+ / Flask 3.1.3
Database	Relational SQL store	SQLite 3 (ACID-compliant, embedded)
Frontend	Modern Web Browser	HTML5, Responsive CSS3, ES6 JavaScript
Deployment Server	WSGI compatible	Gunicorn 26.2+ / Render.com Cloud

5. TESTING & VERIFICATION SUMMARY

The project was verified through **23 automated test cases** covering route availability, FAQ keyword searching, admin login authorization, CRUD lifecycles, and edge-case AI failure degradation. All 23 tests passed with a 100% success rate in 1.185 seconds.

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Student Signature	**Project Guide Signature**
(Akash rana / Shreyash katke)	Prof. [Guide Name]